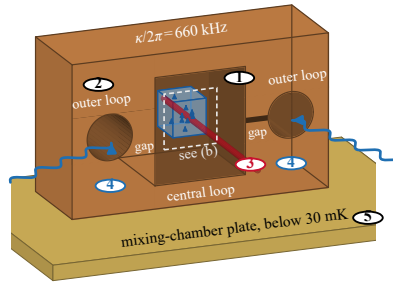
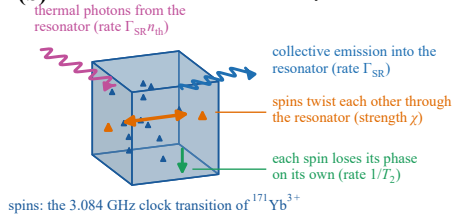


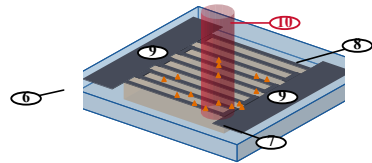
(a) the demonstrated experiment: crystal in a loop-gap microwave resonator



(b) what the model follows inside the crystal



(c) proposed planar superconducting resonator on the same crystal (not built)



Key

- ① $^{171}\text{Yb}^{3+}:\text{CaWO}_4$ crystal, $4.4 \times 4.6 \times 5 \text{ mm}$, $4.96 \text{ ppm } ^{171}\text{Yb}$, glued in the central loop
- ② metal loop-gap resonator: central loop, two outer loops, narrow gaps ($g/2\pi = 15 \text{ mHz}$, $V_{\text{m}} = 275 \text{ mm}^3$)
- ③ 973 nm light through the crystal: optical pumping into $|\downarrow\rangle$ and absorption readout of the populations
- ④ microwave antennas at the outer loops: spin control pulses and transmission measurement
- ⑤ mixing-chamber plate of the dilution refrigerator (spin temperature 80 mK, $n_{\text{th}} = 0.19$)
- ⑥ CaWO_4 chip carrying the resonator
- ⑦ read-out volume under the inductor, where the microwave field is nearly uniform (coupling spread D)
- ⑧ niobium meander inductor (the part of the resonator that couples to the spins)
- ⑨ niobium capacitor pads
- ⑩ optical read-out beam through the read-out volume